Intro to Image Analysis: Part 1

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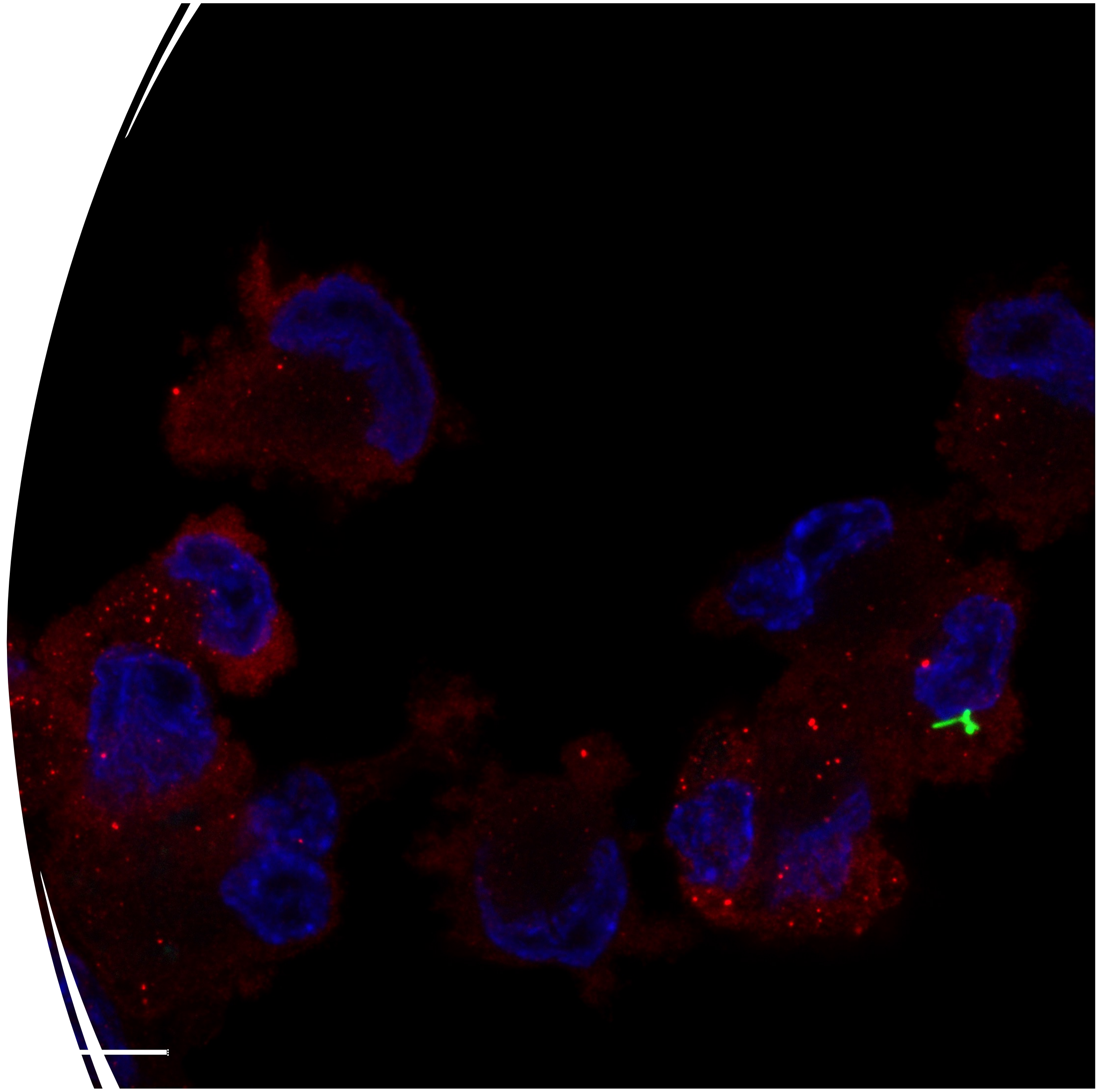
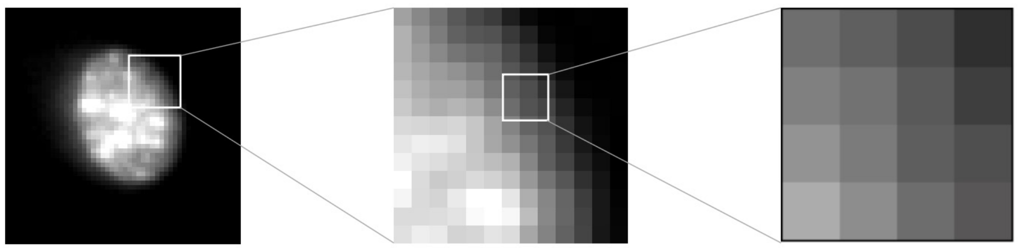


What we’re learning today

- What images really are
- How to work with images
    - It’s a data format many aren’t used to and can be tricky to handle
- Overcoming common roadblocks when getting started with image processing & analysis
- Hands on example – Using densitometry to quantify a western blot

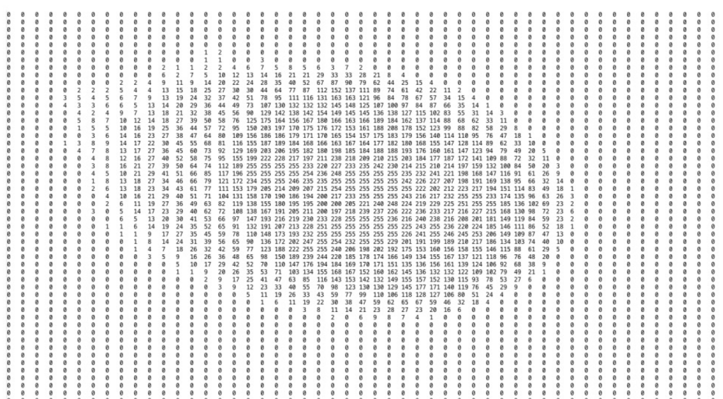
What is an Image (actually)?

An image is an array of numbers representing intensity values



=

110	95	76	47
128	114	89	62
147	123	94	79
172	141	109	88



Images = Pixels (or voxels)
Pixels = Numeric value
Same pixel values can be displayed differently using lookup tables (LUTs)

LUTs

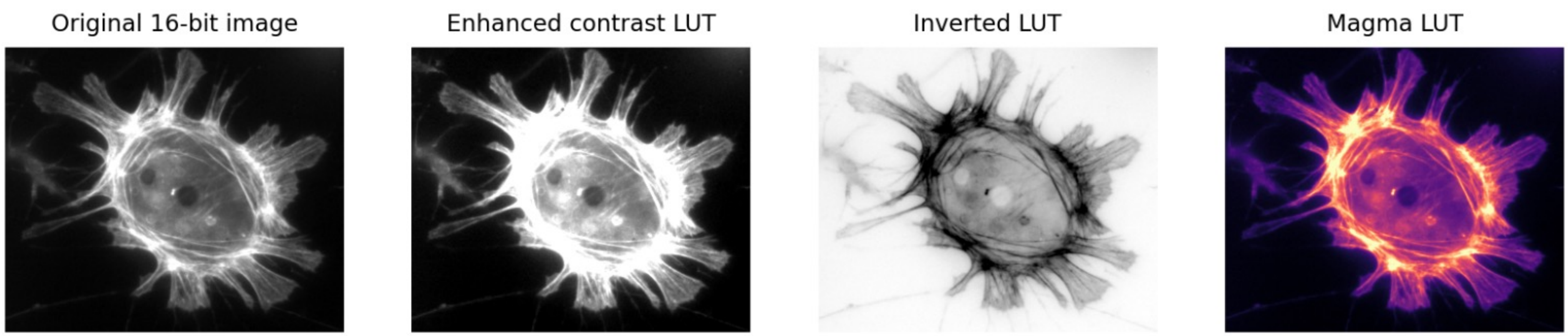


Fig. 5 Images that *look* different, but contain *the same* pixel values. Measuring each of these images would give the same results.

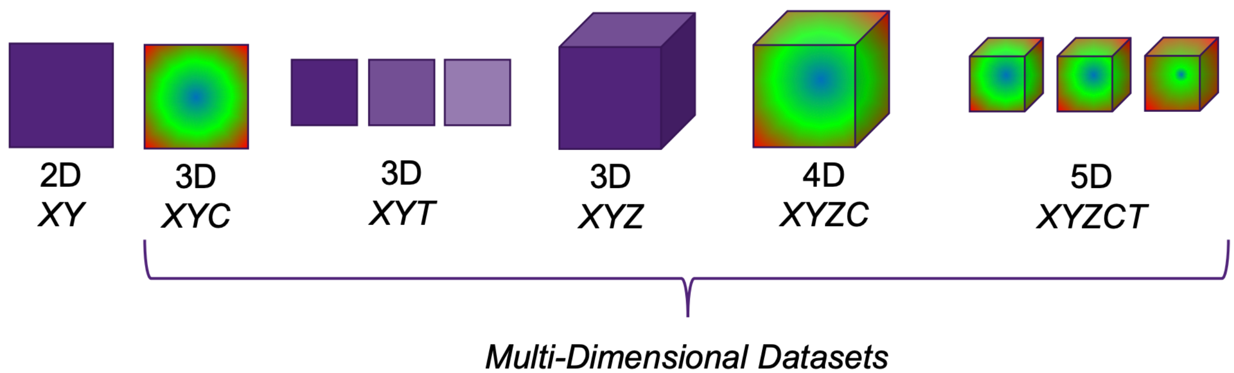
https://bioimagebook.github.io/chapters/1-concepts/1-images_and_pixels/images_and_pixels.html

Image Dimensions

width, height, depth, channel, time

We use standard terms to describe image datasets;

Width = *X*
Height = *Y*
Depth = *Z*
channel = *C*
Time = *T*



**Note that the order a device saves your data into a file may vary from system to system.*

Colors + Channels

RGB images – 3 values stored per pixel for Red, Green, & Blue

for display mainly, not very good for quantitative analysis

‼ *once you convert an image to RGB, you can't go back! It's a common way to lose raw data*

Multichannel/composite images – better for analysis

Uses a separate image for each color

Bit Depth

Bit depth = the number of bits used to represent each pixel

Bit depth imposes a limit on the pixel values that are possible

Lower bit-depth → fewer pixel values possible → image may have less info

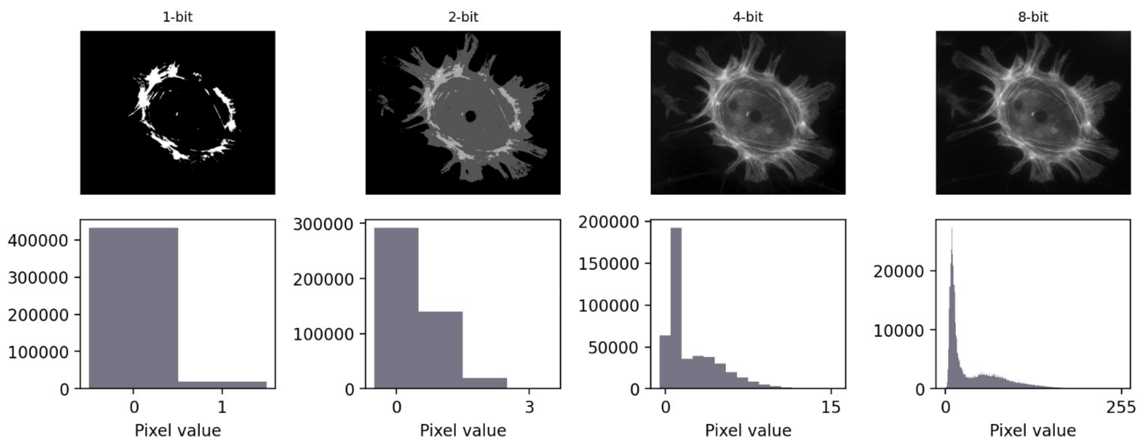


Fig. 21 Representing an image using different bit-depths.

https://bioimagebook.github.io/chapters/1-concepts/3-bit_depths/bit_depths.html

Common Image File Formats
(+ which ones you should avoid in analysis)

There's a lot of ways you can save or export your image files…

Some file formats will mess up your image for further analysis
Or they will lose the image metadata

Best Practice: Always keep your original files, in their original format untouched & safe

Common Image File Formats
(+ which ones you should avoid in analysis)

Format	Extensions	Main use	Compression	Comment
TIFF	.tif, .tiff	Analysis, display (print)	None, lossless, lossy	Very general image format
OME-TIFF	.ome.tif, .ome.tiff	Analysis, Display (print)	None, lossless, lossy	TIFF, with standardized metadata for microscopy
Zarr	.zarr	Analysis	None, lossless, lossy	Emerging format, great for big datasets – but limited support currently
PNG	.png	Display (web, print)	Lossless	Small(ish) file sizes without compression artefacts
JPEG	.jpg, .jpeg	Display (web)	Lossy (usually)	Small file sizes, but visible artefacts

https://bioimagebook.github.io/chapters/1-concepts/6-files/files.html

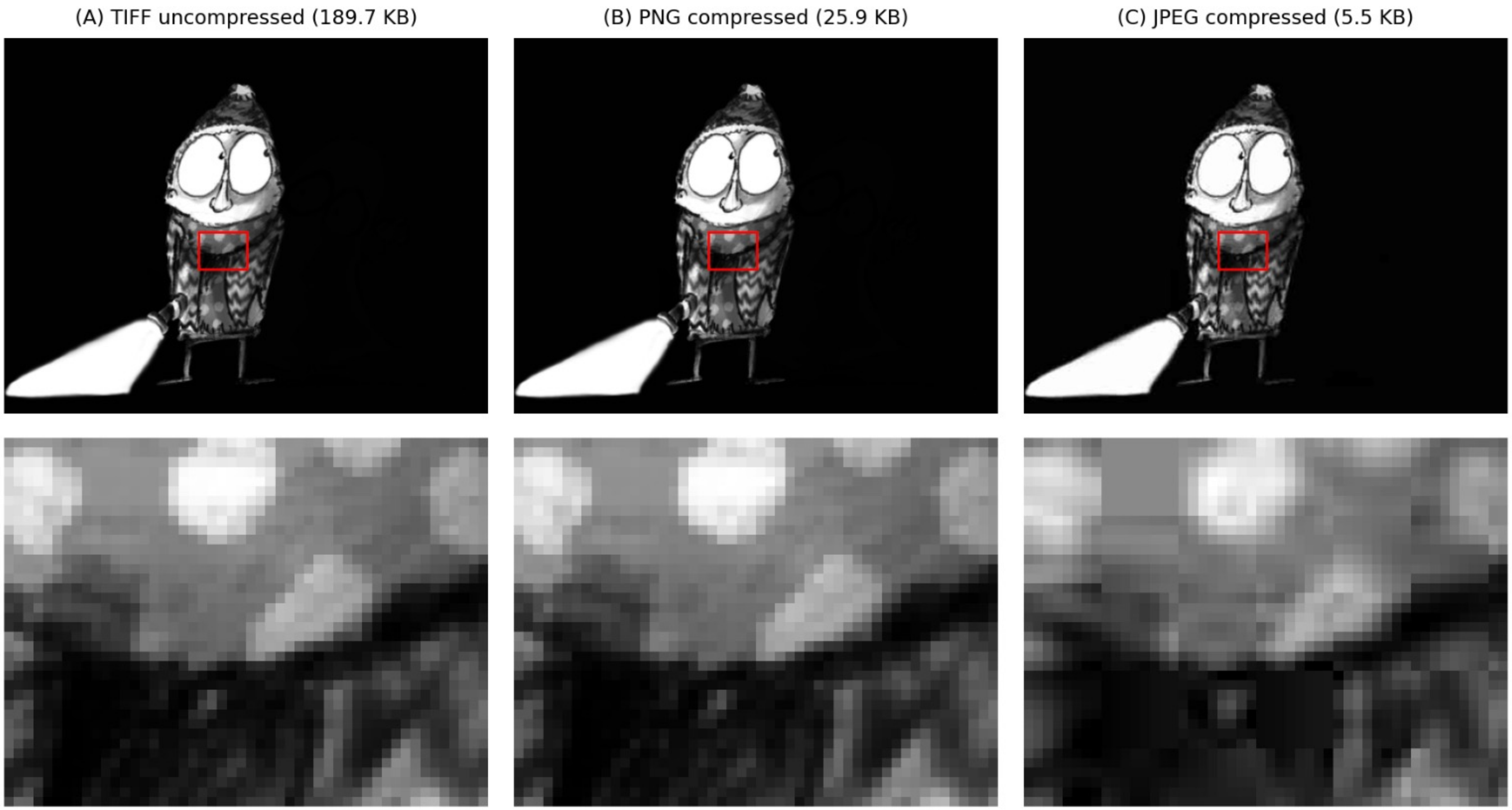
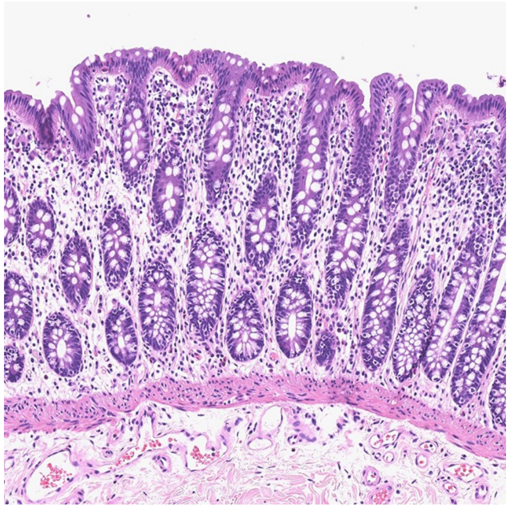


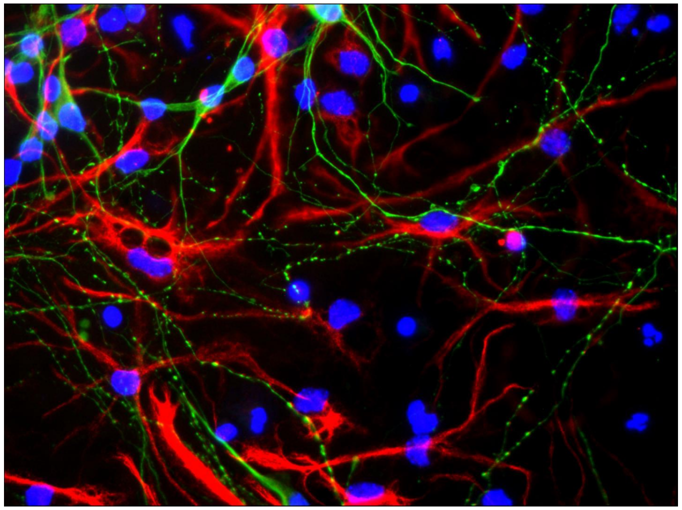
Fig. 53 Examples of images saved with (A) no compression, (B) lossless compression, and (C) lossy JPEG compression. The pixel values of (A) and (B) are identical. Image (C) looks similar, but zooming in on a detailed region reveals characteristic JPEG artefacts. #

Common Science Images

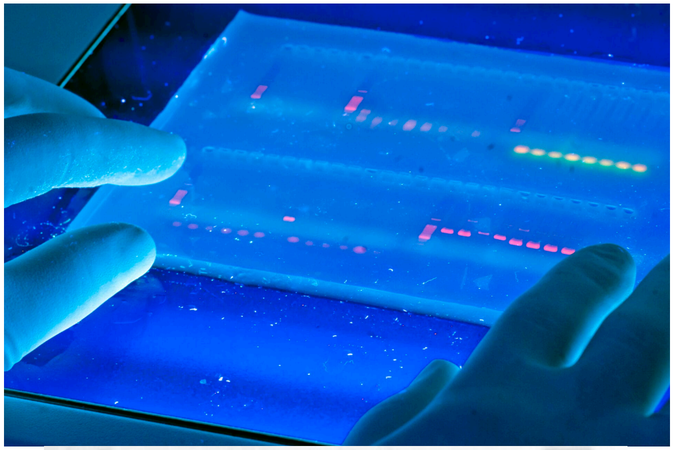
Histology/Pathology

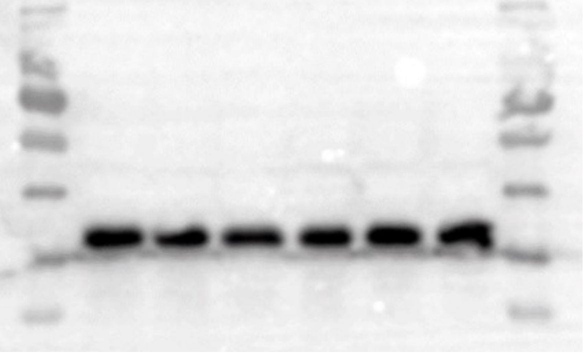


Fluorescent Microscopy



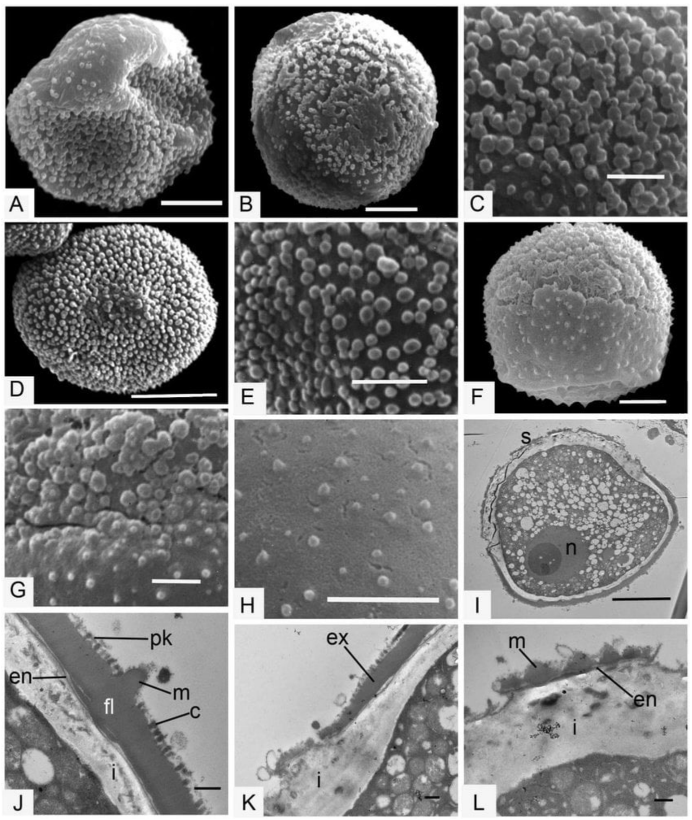
Gel Images



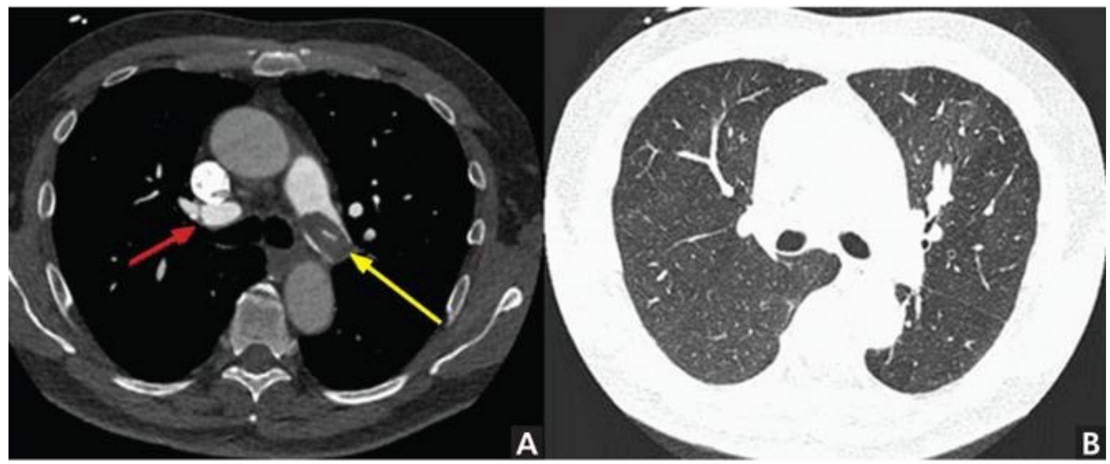


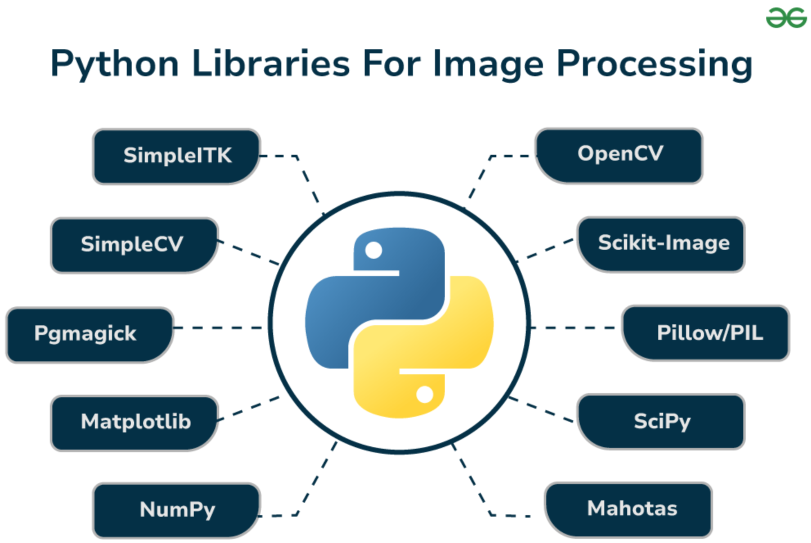
Common Science Images

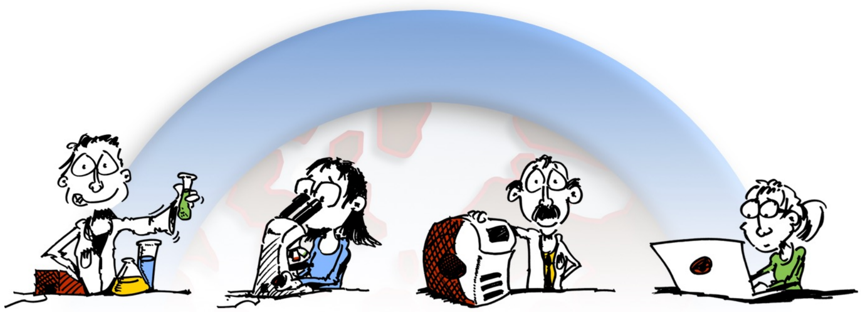
SEM/TEM



Medical Imaging – CT/MRI/X-Ray





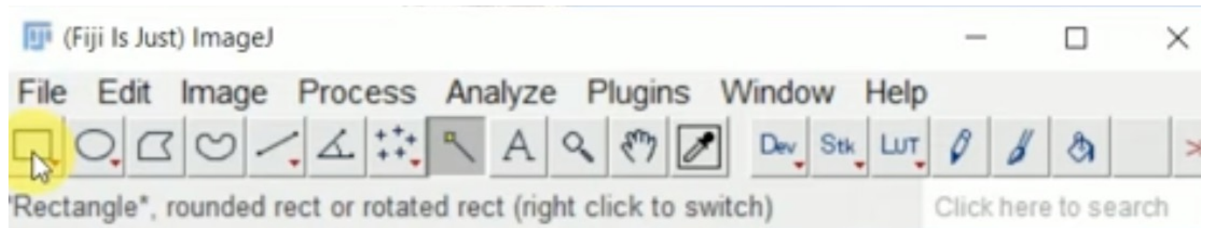


QuPath

Open Software for Bioimage Analysis

VISIOPHARM®

CellProfiler™
cell image analysis software



ImageJ/FIJI

Why am I using these terms interchangeably??

Because F̲IJI I̲s J̲ust I̲mageJ

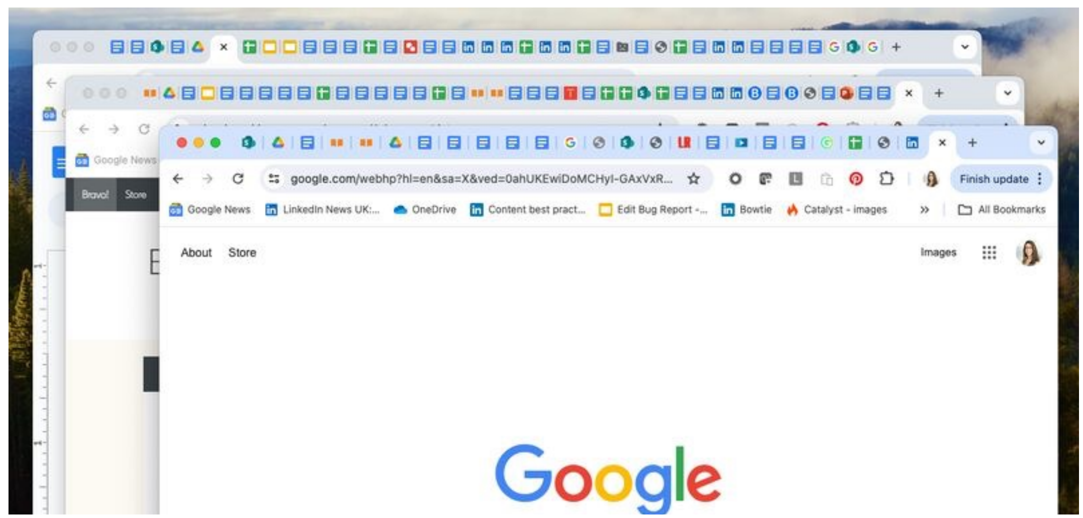
Can everyone open FIJI?

Welcome to ImageJ

- ImageJ is free & open source
- Very widely used & has many many different applications
- FIJI is the "batteries included" version that has functional plugins

It operates in many boxes/windows so if you are this type of person →

You're gonna do great



The ImageJ Interface

ImageJ’s user interface is rather minimalistic. It’s centered around a toolbar. Everything else (images, histograms, measurement tables, dialogs) appears within separate windows.

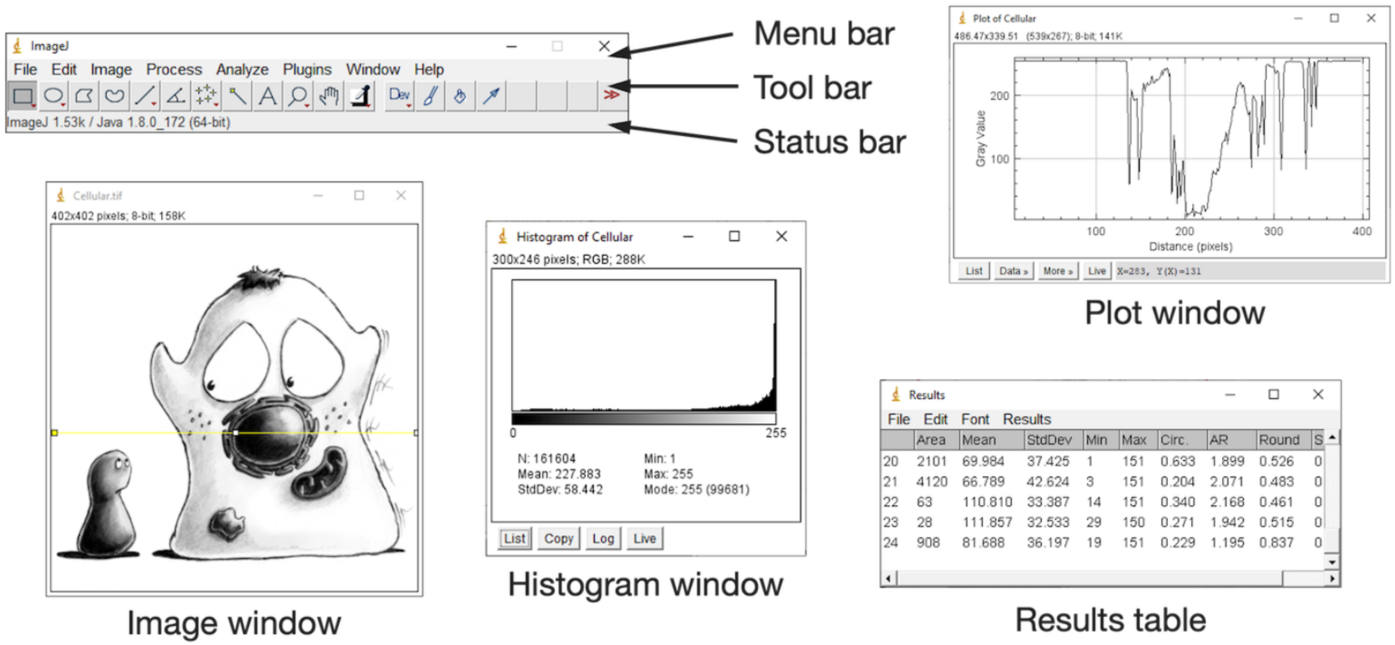
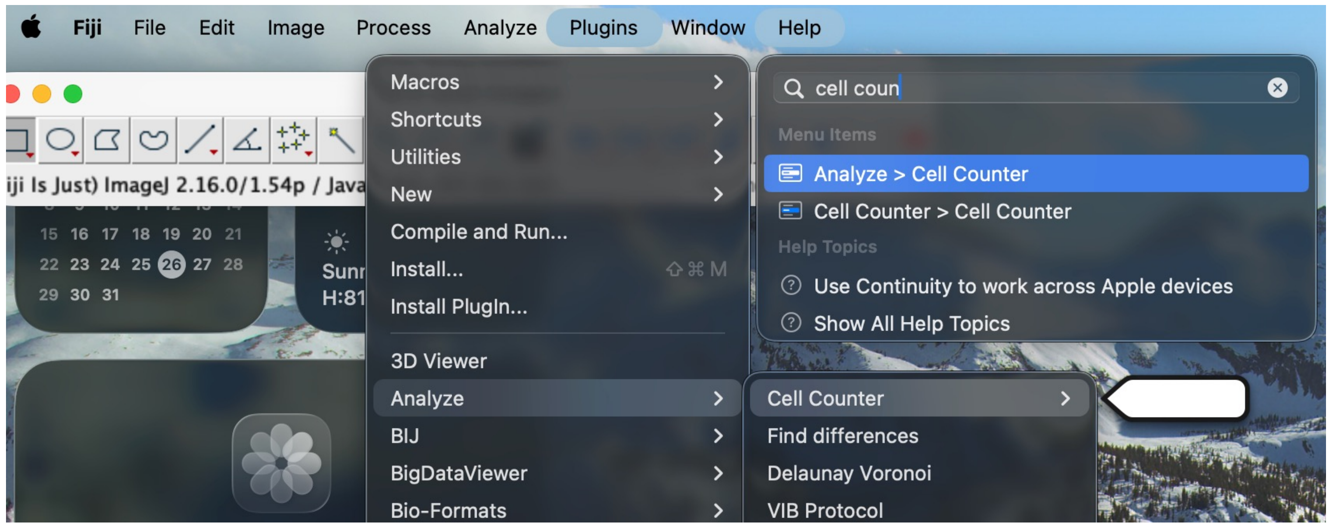


Fig. 6 The main ImageJ user interface.

Pro Tip : Search!



Many functionalities are buried in the menus
But the search tool is extremely helpful as it will show you exactly where the tool is located!

Fun stuff you can do in ImageJ/FIJI!

- Measure!
- Count!
- Draw!
- And much much more!

What does ImageJ provide?

Application	A user interface with functions to load, display, and save images.
Techniques	Image processing, colocalization, deconvolution, registration, segmentation, tracking, visualization and much more.
Plugins	A powerful mechanism for extending ImageJ in all kinds of useful ways.
Scripting	Automated, reproducible workflows via scripts and macros, including headless on a remote server or cluster.
Forum	A vibrant, diverse, and helpful user community that gives rise to insightful scientific exchanges.

Let’s visualize this image as a person with colorblindness would see it!

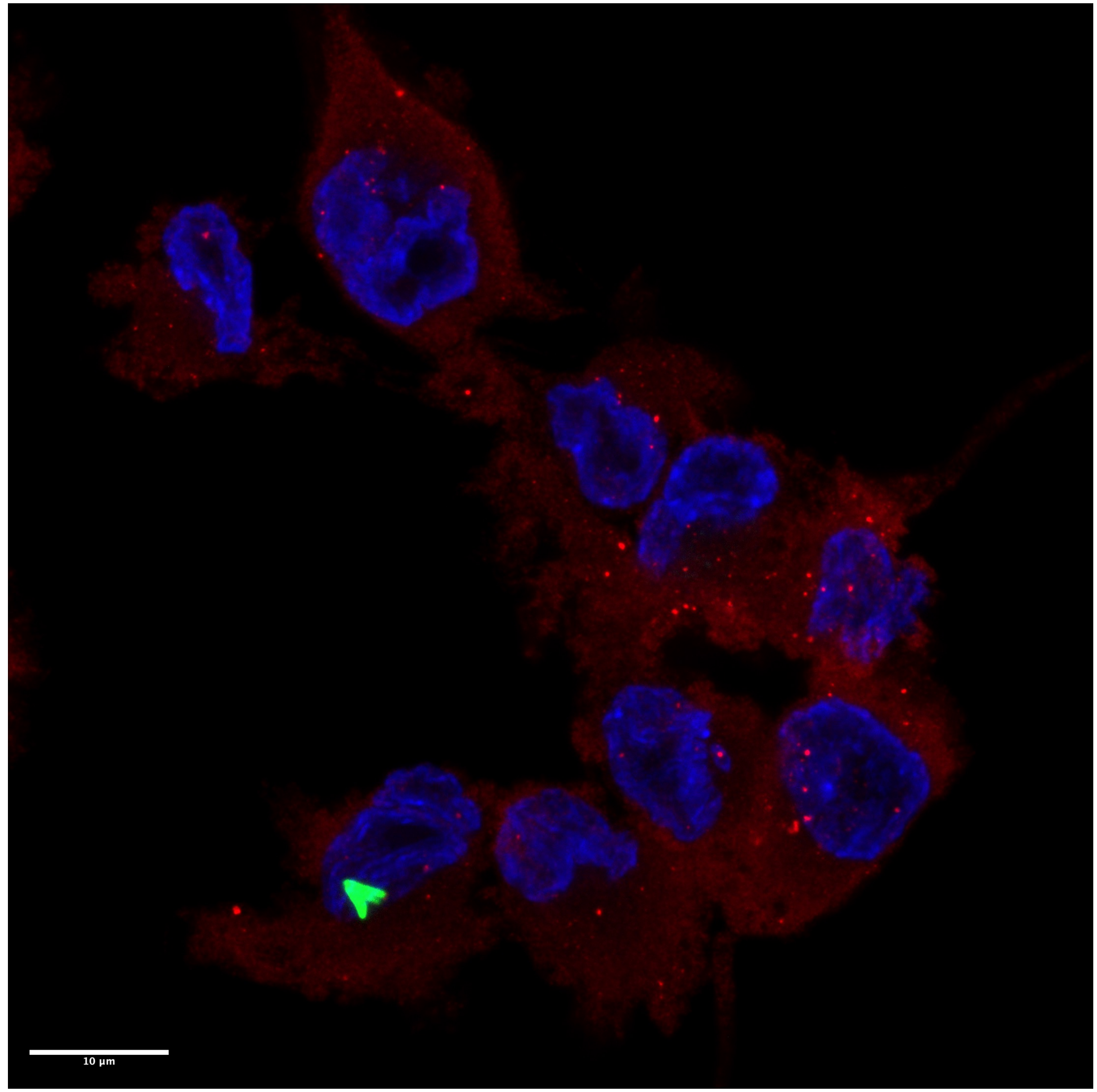
Open Image1.png from Image Bank

Navigate to:

Image > Color > Simulate Color Blindness

Select the type of color blindness from the drop-down menu

Helpful for determining accessibility of your color choices (and fun)



My Image Analysis Journey

Let's go over some of the headaches and roadblocks I faced as I was learning to process and analyze images!

Maybe my trials and tribulations will make your life easier in the future

Where do the microscopes live?

- You probably will not have your own personal microscope
  - So that's why we share <3
- Microscopes in the Microscope Imaging Network (MIN) can be accessed by individuals across campus
  - Each microscope has someone in charge of it, you contact this person, they work with you to arrange training, access, and billing
  - FMIAC in MRB: https://www.bmb.colostate.edu/flourescence-microscopy-image-analysis-center/

$157,500

Zeiss LSM 900 Microscope

Keyence All in One Fluorescence Microscope ~ $20,000

How do I open this image???

Many different formats – often proprietary/specific to microscope

- Lots of info encoded in these (dimensions, pixel size, scale, image getting information like microscope, objective, lens)
  - You should keep a pristine, unedited copy of this in a safe folder somewhere
- Your computer probably doesn’t know how to handle this info
- But luckily, ImageJ has a tool for that! BioFormats!

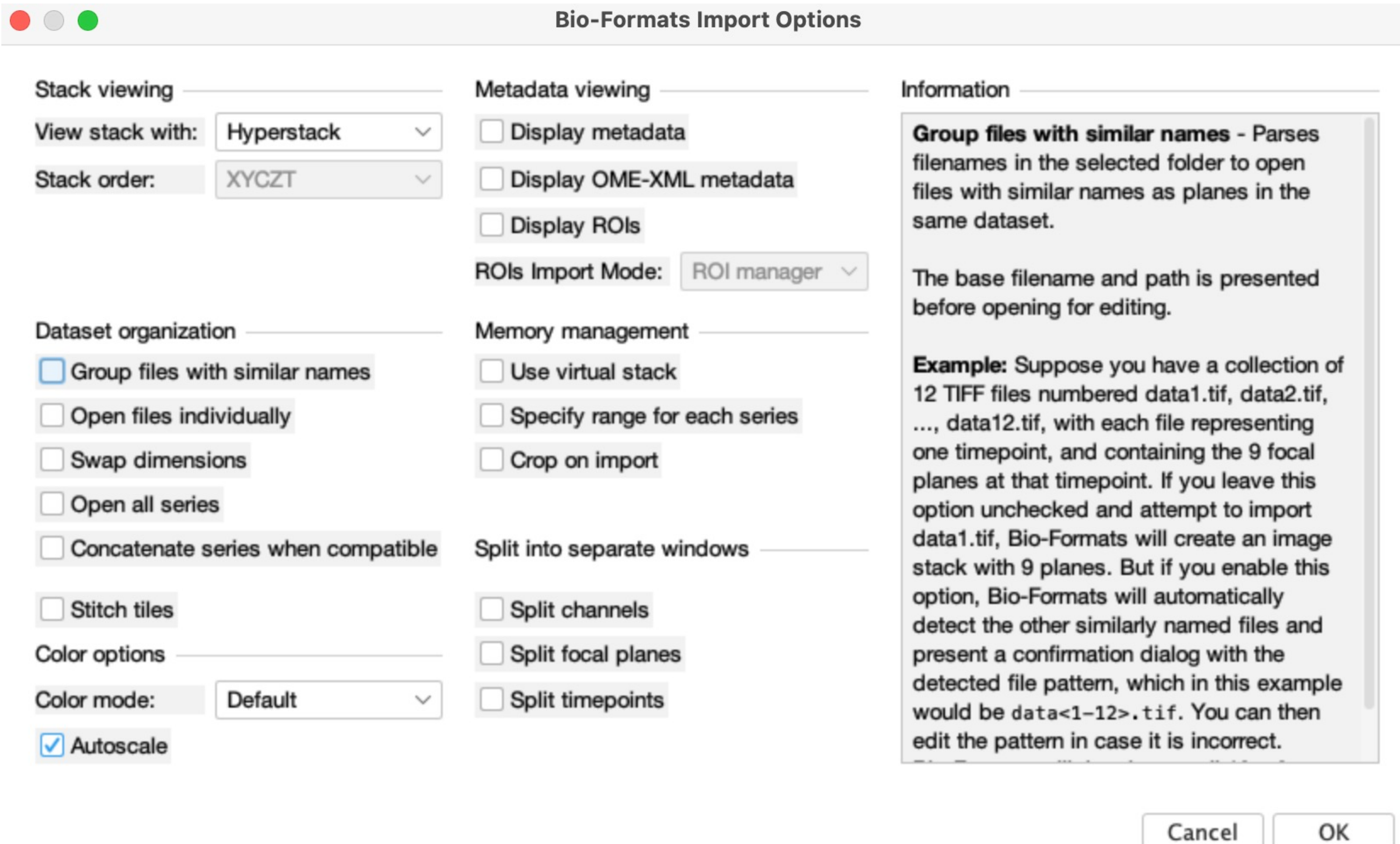
Bio-Formats currently supports **165** formats

Open “Image2.czi” from image bank in FIJI

Drag & Drop file

Ctrl/Cmd + O

File > Open

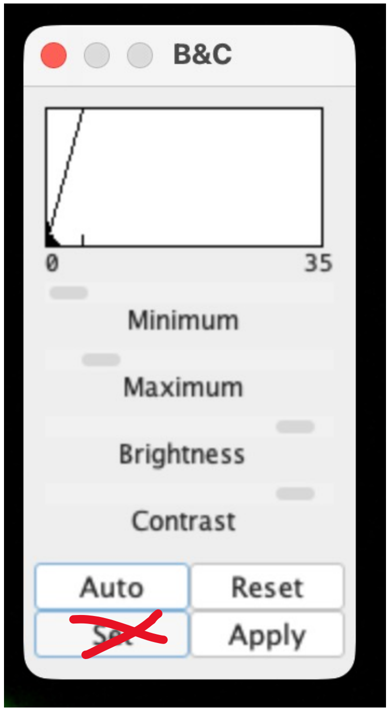

Bio-Formats Import Options
Stack viewing
View stack with:
Hyperstack
Stack order:
XYCZT
Metadata viewing
Display metadata
Display OME-XML metadata
Display ROIs
ROIs Import Mode:
ROI manager
Information
Group files with similar names - Parses filenames in the selected folder to open files with similar names as planes in the same dataset.
The base filename and path is presented before opening for editing.
Example: Suppose you have a collection of 12 TIFF files numbered data1.tif, data2.tif, ..., data12.tif, with each file representing one timepoint, and containing the 9 focal planes at that timepoint. If you leave this option unchecked and attempt to import data1.tif, Bio-Formats will create an image stack with 9 planes. But if you enable this option, Bio-Formats will automatically detect the other similarly named files and present a confirmation dialog with the detected file pattern, which in this example would be data<1-12>.tif. You can then edit the pattern in case it is incorrect.
Dataset organization
Group files with similar names
Open files individually
Swap dimensions
Open all series
Concatenate series when compatible
Stitch tiles
Memory management
Use virtual stack
Specify range for each series
Crop on import
Split into separate windows
Split channels
Split focal planes
Split timepoints
Color options
Color mode:
Default
Autoscale
Cancel
OK

Headache #2: Seeing your Image

- Adjust brightness/contrast (*Image > Adjust > Brightness/Contrast)*
  - Make sure your image box is selected and hit auto

‼Reminder to adjust brightness & contrast equally across all images in a data set (Research Ethics)… We will talk about macros later, these can do it efficiently!
‼Also be sure to use scientific software for scientific images! A lot of other image handling software can't handle 16 or 32 bit images, and are more likely to permanently change the pixel values, spoiling your raw data

This adjustment in ImageJ changes only the LUT, and does not change the pixel values…Unless you hit "Set" (please don't!)



Headache #3: Making your image presentable

Your raw data is super important for the analysis step… but your PI probably wants something pretty to look at/something they can at least open on their computer

Lots of different image manipulations possible in ImageJ

- Adjust contrast/brightness
- Maximum intensity projections
- Add scale bar

It's too much to do all this image by image…

ImageJ Macros

Intro to the scripting part of ImageJ!

Open “make-images-pretty.ijm” in FIJI

Run > Select “Image3.czi”

This should then run & save a presentable, RGB display copy of your image

This macro was for 1 individual image but can be modified to loop thru a file list in a folder

Headache #4: Turning a picture into numbers

- Much of the time we focus on making numbers into pictures (making charts, plots, graphs, diagrams)
  - Let’s do the reverse now!
- Exercise 1

Open your image in FIJI

- Select “WesternBlot1.tiff” from Image Bank folder
- Drag & Drop into box
- *File* > *Open*
- CMD/CTRL + O

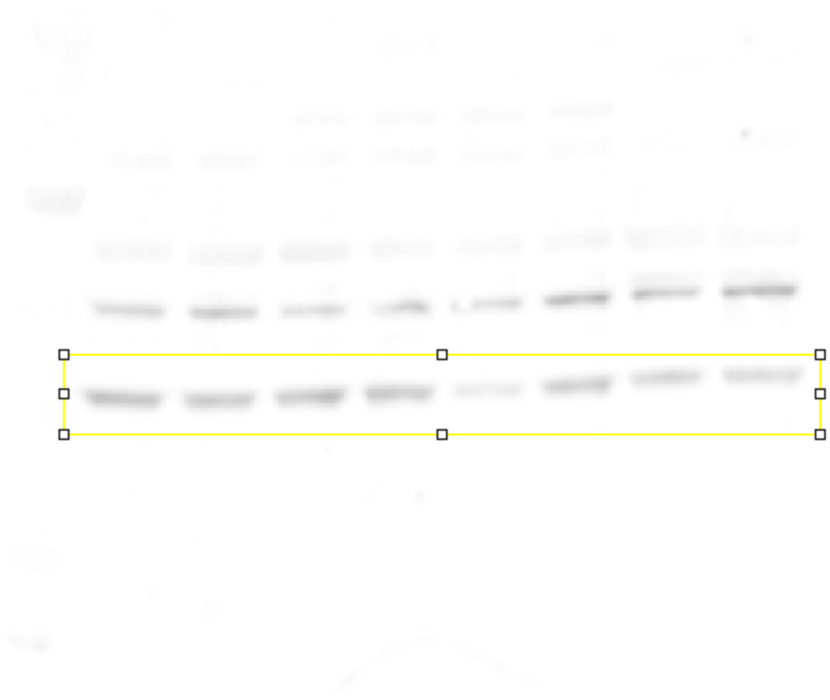
Prepare your image for analysis

- Rotate image so bands are straight
  - *Image > Transform > Rotate*
  - Alternatively, make a rotated rectangle (right click, select rotated rectangle)
    - *Edit > Selection > Straighten*

- Subtract Background
  - *Process > Subtract Background*
  - Since this image has an uneven background and small objects of interest, decrease the pixel size of the rolling ball radius to 10

Select the bands of interest

Bands of interest = bottom darker bands



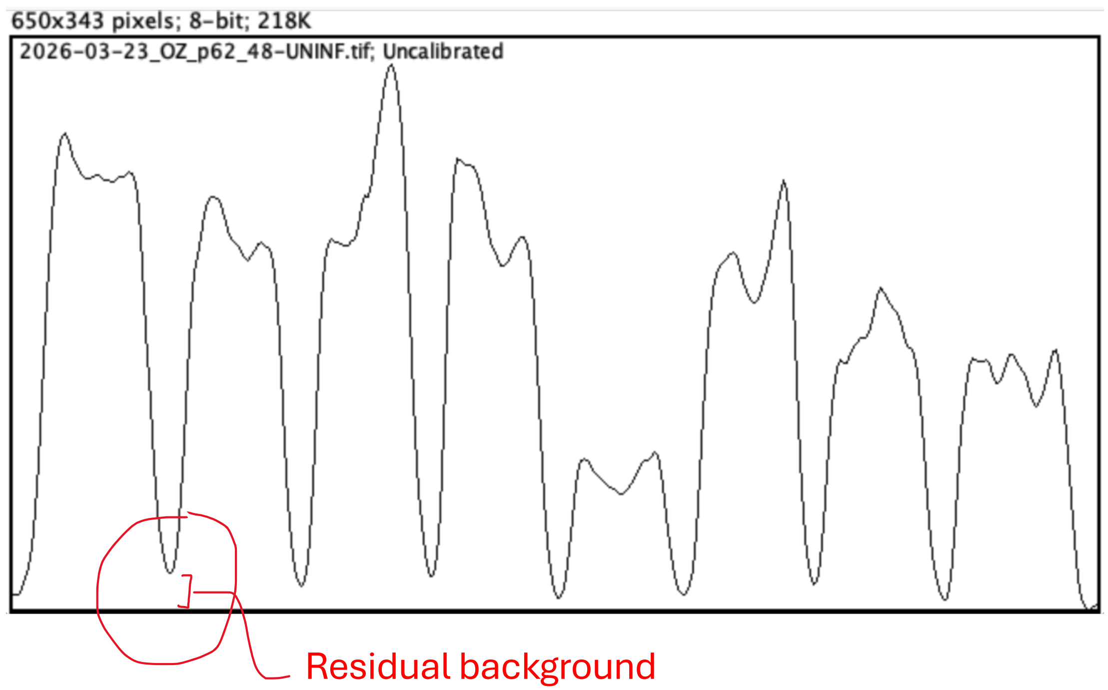
Analyze > Gels > Select First Lane
OR keyboard shortcut
CMD/CTRL + 1

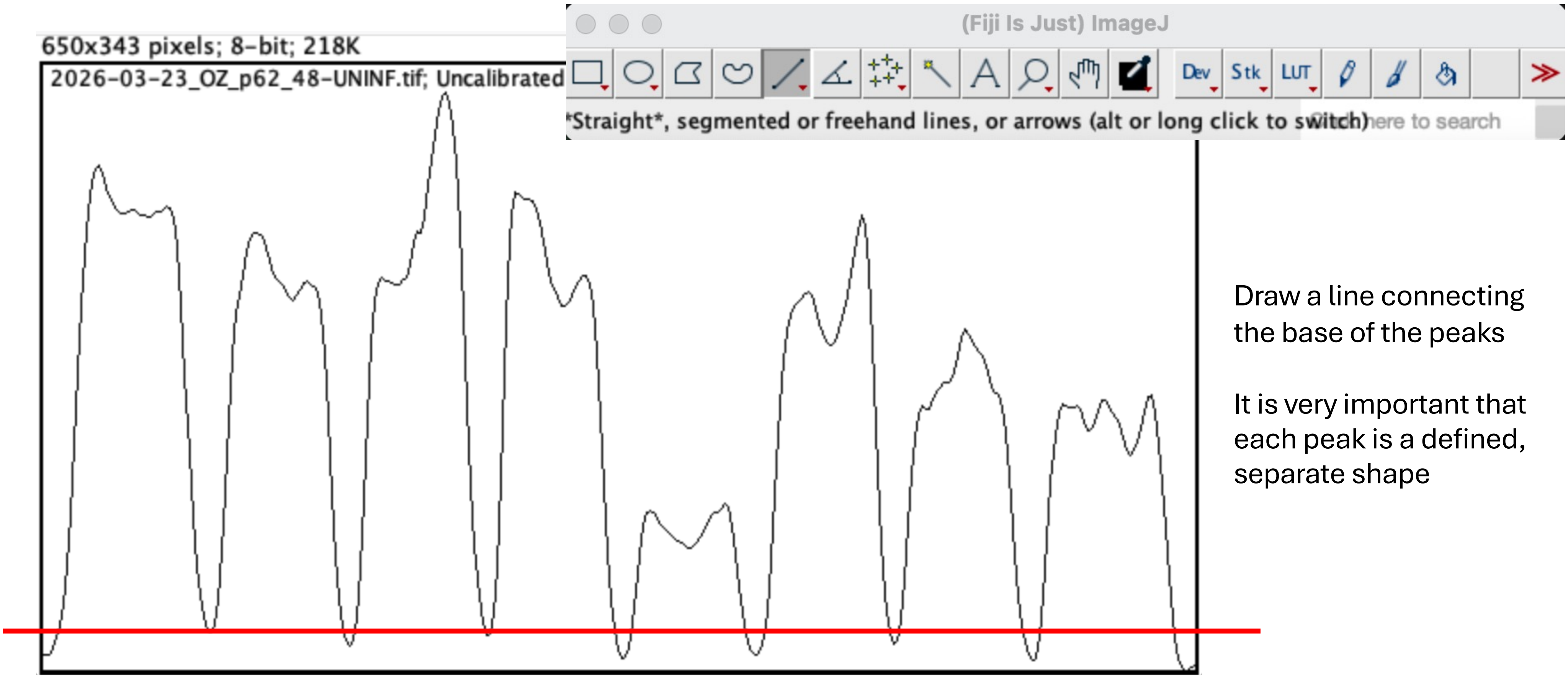
Next,

Analyze > Gels > Plot Lanes

Quantify Bands

- Find the area under the curve of each band of the plotted lane
- First, account for residual background:

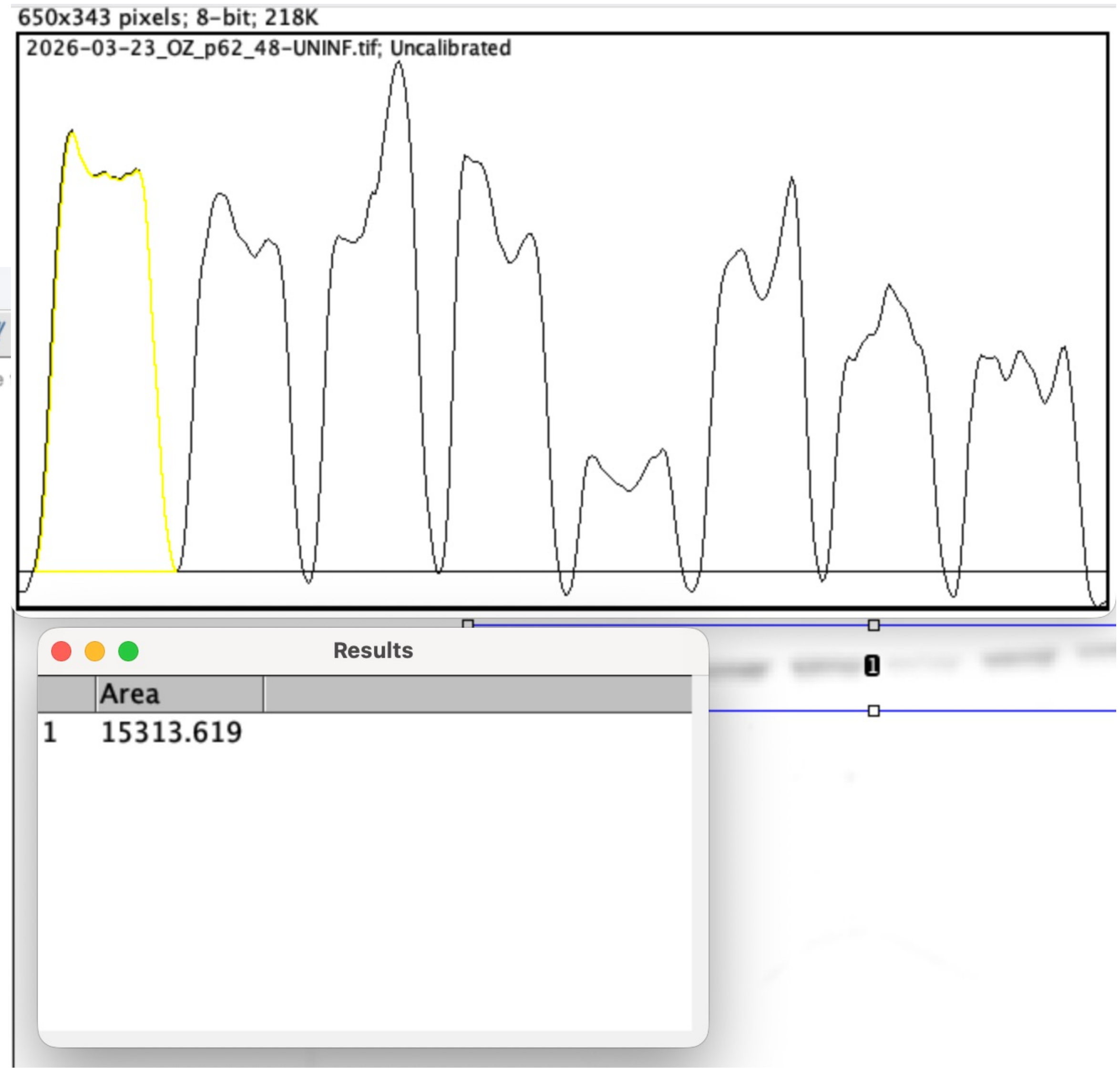




Draw a line connecting the base of the peaks

It is very important that each peak is a defined, separate shape

Next, use the Magic Wand to find the area under each curve by clicking on or in it:

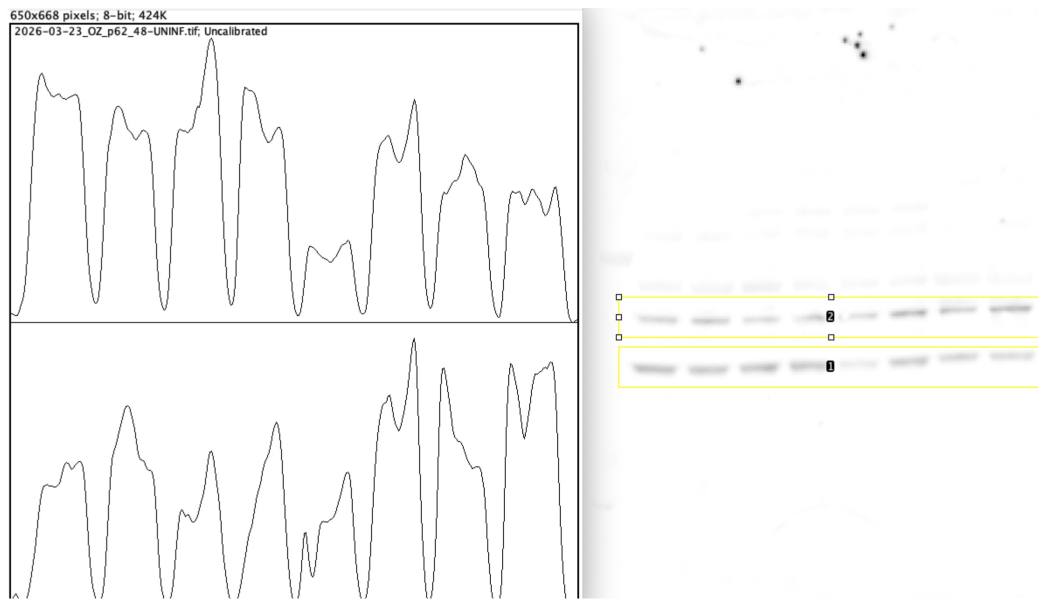


After you've clicked on all the curves, right click on the results and copy paste them into a spreadsheet & plot

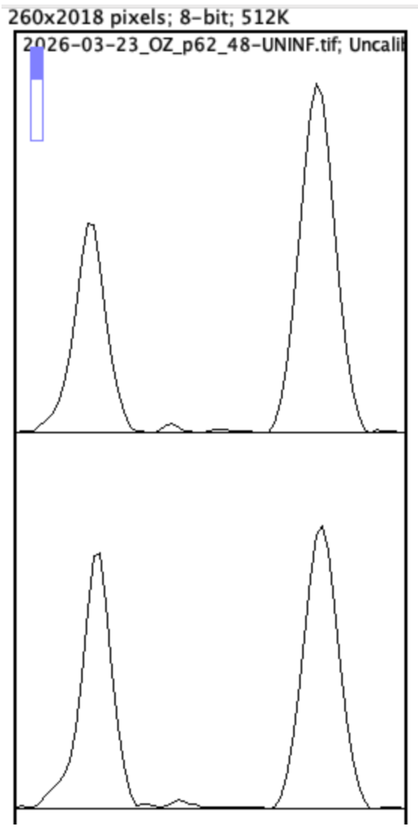
Now repeat for the loading control blot (Western1_LoadingControl.tif), normalize the values to total protein, and then you've analyzed your blot!

What if I want to analyze more than one row of bands on the same gel?

Add another lane:
- Draw a rectangle for the first lane
- Select the First Lane (*Analyze>Gels>Select First Lane)*
- Lane is now labeled with 1.
- Select this box, drag it up to the other band, then select the next lane (*Analyze>Gels>Select Next Lane*)
- Then plot lanes
- Quantify area under each curve for the new row of bands



Also possible to do this with vertical lanes instead of horizontal (it just takes more time!)



More on turning a (more complicated) picture into numbers with Rosi next time!

Image Databases:
Allen Institute of Cell Science
imagescience.org

Python Image Analysis Resources
https://bioimagebook.github.io/chapters/1-concepts/1-images_and_pixels/python.html

UQ – Bio Summer School

ImageJ/FIJI Resources:
Help forum: https://forum.image.sc/
Image J wiki: https://imagej.net

UQ Bio Image Analysis online trainings